Week of 7th Nov.

Tuesday, 7 November 2023 9:43 AM

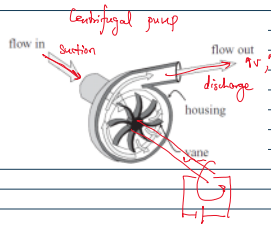
Pump Characteristics Curves

Pumps → mechanical devices that allow for fluid transportation via pipes (Section 9.2.4.1)

↳ Centrifugal pumps are a common type → transfer rotational energy (from electric motor) to fluid movement

+ Working principle is energy balance! Electric energy → Shaft work (by pump to fluid) → affects speed, pressure & other fluid characteristics

+ Pump manufacturers need to characterise a pump → Pump operation under various conditions [Pump curves!]



$\Delta E = Q + W$ (total energy of system; Heat from environment to the system via boundary; Work done on system by environment)

Energy balance on CV: $\frac{dE_{cv}}{dt} + \iint_{CS} (\hat{n}\cdot\vec{v})\rho\hat{E}\,dS = \dot{Q}_{in,cv} - \dot{W}_{s,cv}$ (In, out) $\quad [E_{cv} = U + E_K + E_P]$

Closed system

Open system (mass flow in & out)

Some assumptions

$\Delta E_p + \Delta E_K + \Delta U = Q_{in,cv} - W_{s,by\,cv}$ $\quad (W_s = F\cdot d = (P\cdot A)\cdot d)$

(W_s) Shaft work (work performed by fluid due to moving parts)

Flow work (W_f) (work done by fluid as it enters & leaves the system)

$H = U + PV$

SS Macro Energy balance

$$\Delta E_p + \Delta E_K + \Delta H = Q_{in,cv} - W_{s,by\,cv}$$

Single in, single out, Incompressible fluid

$$\dot{m}\left(\frac{\langle v\rangle_2^2}{2} - \frac{\langle v\rangle_1^2}{2}\right) + \dot{m}g(z_2 - z_1) + \dot{m}\left(\hat{U}_2 - \hat{U}_1 + \frac{P_2}{\rho} - \frac{P_1}{\rho}\right) = \dot{Q}_{in,cv} - \dot{W}_{s,by\,cv}$$

Temp constant, no phase changes/chemical reaction, negligible heat transfer

Mechanical energy Balance (MEB)

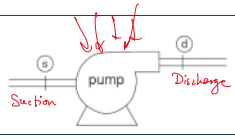
pressure

$$\frac{\Delta P}{\rho} + \frac{\Delta\langle v\rangle^2}{2\alpha} + g\Delta z + F = -\frac{W_{s,by}}{m}$$

density; average velocity; $\alpha = 1$ (turbulent flow); friction term $[\Delta U - Q]$

Pump can add energy via:
+ Increase velocity of fluid
+ Create pressure differential
+ Pump can change fluid height (raise its potential energy)

PFD



η = Pump efficiency = $\frac{\text{(useful energy)}}{\text{(total energy)}}$ (energy converted to work (rather than friction))

$$\eta = \frac{\left(\frac{-W_{s,by}}{m} - F\right)}{\left(\frac{-W_{s,by}}{m}\right)}$$

$\eta \left(\frac{-W_{s,by}}{m} \right)$

Pump brake horsepower: $bhp = -W_{s,by}$ (Total work delivered by pump)

↓ Dividing by g

$$\frac{\Delta p}{\rho g} + \frac{\Delta \langle v \rangle^2}{2\alpha g} + \Delta z = \eta \left(\frac{-W_{s,by}}{mg} \right) = \eta \cdot (bhp)$$

out in $\eta \cdot bhp$

$$\left(\frac{P_d}{\rho g} + \frac{\langle v \rangle_d^2}{2g} + z_d \right) - \left(\frac{P_s}{\rho g} + \frac{\langle v \rangle_s^2}{2g} + z_s \right) = \eta \left(\frac{-W_{s,by}}{mg} \right)$$

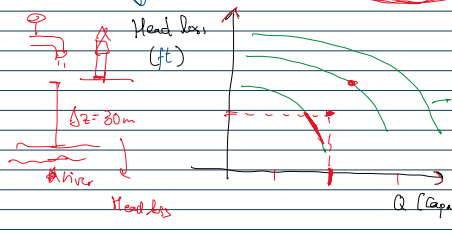
Discharge head (H_d) [=] length

Suction head (H_s) [=] length

$$(H_d - H_s) = \eta \left(\frac{-W_{s,by}}{mg} \right) = \eta \left(\frac{-bhp}{mg} \right)$$

Head loss

give info about η & bhp



Pumping Ability of a CP:
+ Shaft rotational speed (RPM)
↑ Shape of vanes
+ flow path

Lab: Characterize relationships b/w discharge flow rate, net head, bhp, efficiency in a CP

↳ Vary Q (via rotameter)
↳ Measure pressure loss → calculate head loss (H_{loss} or $H_{net} = H_d - H_s$)
↳ Non-dimensionalize variables & then plot them!

Δz, $\langle v \rangle$

PROBLEM: 8.30 Calculate the extra stress tensor $\underline{\underline{\tau}}$ for potential flow around a sphere of an inviscid, Newtonian fluid. Calculate the total stress tensor $\underline{\underline{\Pi}}$. Comment on what is obtained.

$\mu = 0$; $Re \to \infty$

relativistic

$\gamma_{lg} = 0$ air

liquid/interface

flow

∇P

$$\tau = \mu \left((\nabla \underline{v}) + (\nabla \underline{v})^T \right)$$

$$\rho \left(\frac{\partial \underline{v}}{\partial t} + (\nabla \underline{v}) \cdot \underline{v} \right) = -\nabla P + \mu (\nabla^2 \underline{v}) + \rho \underline{g}$$

nondim

$\frac{1}{Re} (\nabla^2 \underline{\hat{v}})$

$Re \to \infty$

$\underline{\underline{\tau}} \approx 0$ ✓

Dynamic pressure

$\nabla P = \nabla p + \rho g$

$$\underline{\underline{\Pi}} = \underline{\underline{\tau}} - p \underline{\underline{I}}$$

PROBLEM: 8.40 What is the force (i.e., drag) on a thin plate given the following conditions? The fluid is water, the plate is 0.52 m long and 0.3 m wide, and the free-stream velocity is 1.3×10^{-2} m/s.

Boundary layer flow:

$\nabla \cdot \underline{v} = 0$

Navier Stokes

v varying in two dimensions

Assumptions

Velocity profile for laminar flow past a flat plate

$$\frac{v_1}{v_\infty} = f(x_2) = 1 - (0.5434)\log\left[1 + (68.3)10^{-0.624 x_2}\right]$$

$$\underline{F} = \iint_{surface} [\hat{n} \cdot \underline{\underline{\Pi}}]\, dS$$

$\underline{\underline{\Pi}} = -p\underline{\underline{I}} + \underline{\underline{\tau}}$

$\underline{\underline{\tau}} = \left(\frac{\partial v}{\partial n}\right) \ldots$

$\tau_w = 0.332 \mu v_\infty \left(\frac{\rho v_\infty}{\mu x_1}\right)^{1/2}$

Drag is the force component in flow direction

